

# Predicting Hypotensive Events with Transformer: A Comparative Performance Analysis

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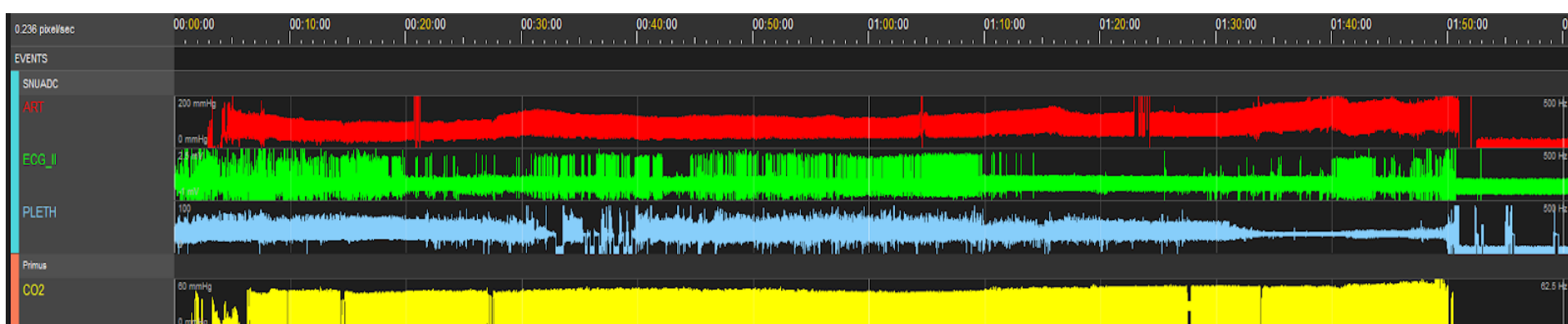
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## Abstract

We successfully replicated the paper *Deep Learning Models for Predicting Intraoperative Hypotension*, which utilize a CNN model to predict hypotensive events in the preceding 5, 10, and 15 minutes, both as classification and regression tasks. After replicating the paper, we transitioned from CNN to utilizing a transformer model for hypotension event prediction. Moreover, in this phase, we aim to uncover the advantages of using a transformer model in this application.

## Method

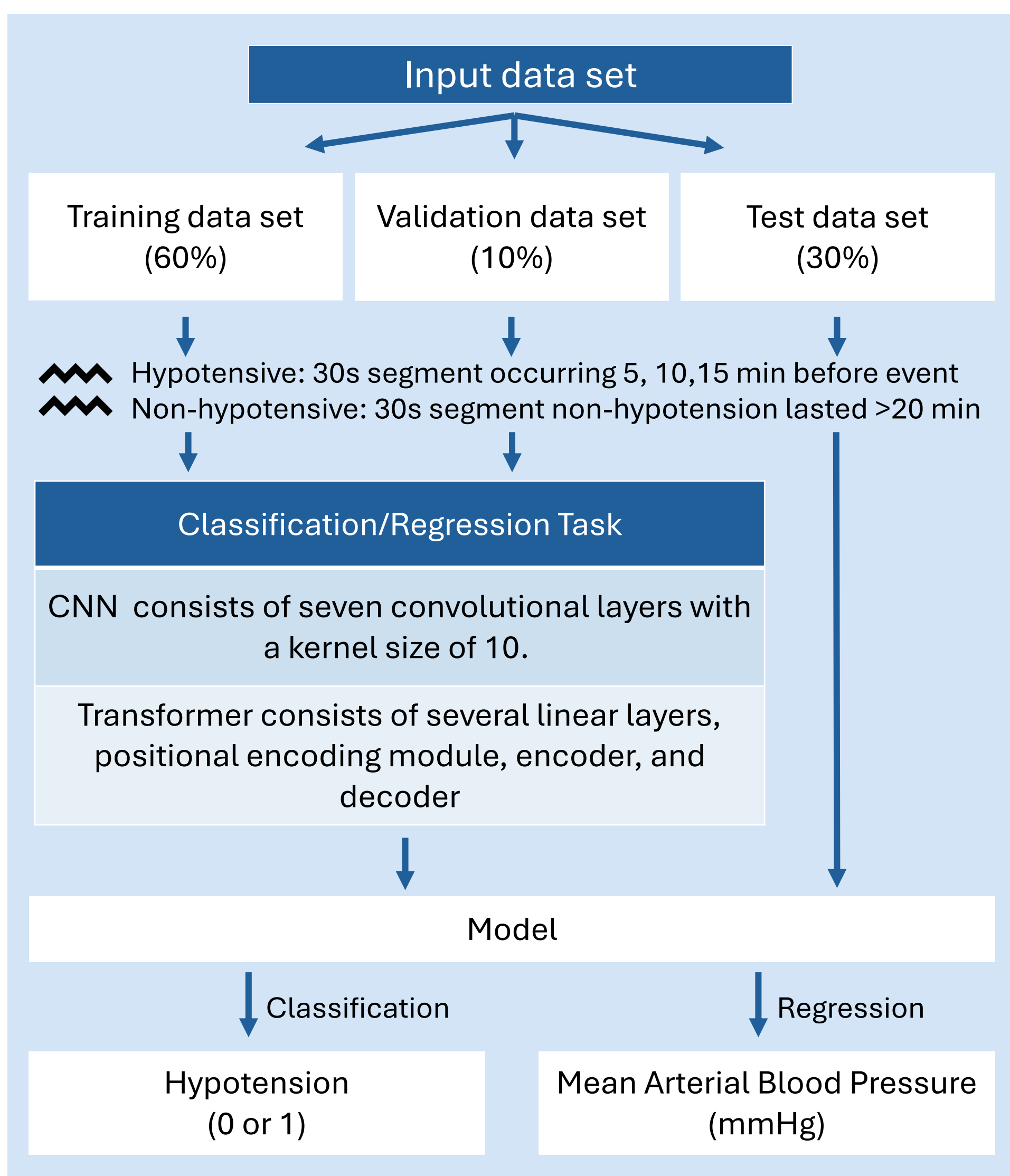
**Dataset:** Four bio-signal waveforms downloading from VitalDB, including Arterial blood pressure wave (ART), ECG lead II wave (ECG\_II), Photoplethysmography wave (PLETH), Capnography wave (CO2).



**Model Types (according to input data set):**

- Invasive input: ART only
- Invasive multichannel input: ART, ECG\_II, PLETH, CO2
- Noninvasive input: PLETH only
- Noninvasive multichannel input: ECG\_II, PLETH, CO2

**Model Building:**



## Next Plan

After comparison, we found that using the transformer model can yield slightly better results than CNN, and its performance remains unaffected as time progresses. Therefore, we aim to make predictions over longer time spans and compare the performance differences between CNN and the transformer model.

## Current Result

